

Due on monday, October 9th, in the morning class.

The questions indicated as **[SELF]** are for self-study and additional exercise.

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1. (a) **[SELF]** Derive, using ‘integration by parts’, the result

$$\int_0^\infty du \, u \, e^{-bu} = \frac{1}{b^2}.$$

You might need this integral later on in this assignment.

- (b) **[SELF]** Teach yourself how to use a computer algebra program (e.g., MATHEMATICA, MAPLE or MAXIMA) for calculating definite and indefinite integrals. Or: how to use an online integrator (search on google for ‘online integrator’). Check the result above.

2. Normalization. Assume the constants α_i to be real and positive.

- (a) **[8 pts.]** Consider a single particle confined to the half-line $x \geq 0$. The inner product is given by $\langle \phi | \xi \rangle = \int_0^\infty \phi^*(x) \xi(x) dx$. Find the constant α_1 so that the wavefunction

$$\psi(x) = \alpha_1 \sqrt{x} e^{-x}$$

is normalized.

- (b) **[3 pts.]** Consider a spin-1/2 (two-state) system. Find the constant α_2 so that the following state is normalized:

$$|\psi\rangle = \alpha_2 \begin{pmatrix} 2i \\ -1 + i \end{pmatrix}.$$

- (c) **[7 pts.]** Consider a system with an infinite-dimensional Hilbert space, for which any state can be expressed as a linear combination of the set of states $\{|\phi_n\rangle\}$, with $n = 0, 1, 2, \dots$. This set of ‘basis states’ satisfy $\langle \phi_m | \phi_n \rangle = \delta_{mn}$. Find the constant α_3 so that the following state is normalized:

$$|\psi\rangle = \alpha_3 \sum_{n=0}^{\infty} \frac{(-1)^n}{3^n} |\phi_n\rangle.$$

- (d) **[4 pts.]** For a system with a 4-dimensional Hilbert space, find the constant α_4 so that the following wavefunction is normalized:

$$|\psi\rangle = \alpha_4 \begin{pmatrix} -2 \\ -1 + 3i \\ i \\ -i \end{pmatrix}.$$

- (e) [8 pts.] For a single particle confined in the x - y plane, find the constant α_5 so that the following wavefunction is normalized:

$$\psi(x, y) = \alpha_5 e^{-(x^2+y^2)/2}.$$

Note: for this system the inner product is given by

$$\langle \phi | \xi \rangle = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \quad \phi^*(x, y) \xi(x, y).$$

3. Discrete energy values and blackbody radiation:

- (a) [10 pts.] In thermal equilibrium, the probability of a mode having energy E is proportional to $e^{-E/k_B T} = e^{-\beta E}$. Here we have defined $\beta = k_B T$.

Show that the average energy of the mode is $k_B T$ or $1/\beta$ if the energy E is a continuous variable, and is $hf/(e^{\beta hf} - 1)$ if the energy is discrete and quantized in units of hf .

You might have to use the summation formulae

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{and} \quad \sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2} \quad \text{for } |x| < 1.$$

- (b) [5+5 pts.] Planck obtained the correct blackbody radiation spectrum by replacing the classical expression by the quantum expression:

$$\frac{1}{\beta} \longrightarrow \frac{hf}{e^{\beta hf} - 1}.$$

Show that the quantum expression reduces to the classical expression at small frequencies. Show this in two ways:

- by taking the limit $f \rightarrow 0$ of the quantum expression.
(You might have to use L'Hopital's rule.)
- by expanding the exponential in the denominator to lowest order.

- (c) [SELF] Express the quantum expression above as a power series in f . The first term should be the classical expression. You could try working out the Taylor series by taking successive derivatives of the full expression (quite tedious: L'Hopital's rule needed multiple times), or you could try expanding only $e^{\beta hf}$ and then using $(1+x)^{-1} \approx 1 - x + \frac{1}{2}x^2 - \frac{1}{3}x^3 + \dots$.